Josh Ederington
ECO 442

Problem Set 3 - Asset Model (Short Run)

1) Congratulations! You have just been appointed the governor of the Banco Central de Costa Rica. For all of the following questions, assume the short-run (i.e., prices are fixed).

a) You have decided that you want to **increase interest rates** in Costa Rica in the **short run** Would you increase or decrease the supply of Costa Rican colóns? To explain, draw the picture for the Costa Rican money market and demonstrate how the change in money supply affects the interest rate.

b) Draw the IRP relationship between the Colón/USD exchange rate and rates of return on investments for a Costa Rican investor (between investments in the U.S. and Costa Rica). Illustrate the effect of a rise in Costa Rican U.S. interest rates on your picture and explain what will happen to the exchange rate of the colón in the **short-run** (i.e., will it appreciate or depreciate versus the dollar?). Hold the expected future exchange rate constant.

c) Explain the impact of this money supply change on the probability that a Costa Rican consumer purchases a car produced in Costa Rica. Assume this decision is a function of (1) the price of the car in Costa Rica (in colons) and (2) the financing the Costa Rican consumer can receive.

d) Explain the impact of this money supply change on the probability that a U.S. consumer consumer purchases a car produced in Costa Rica and then sold in the U.S. Assume this decision is a function of (1) the price of the car in the U.S. (in dollars) and (2) the financing the U.S. consumer can receive.

2) Now assume that an increase in income in Costa Rica has resulted in an **increase in money demand** (for the colón). Draw the Asset model relationship between the Costa Rican Money Market and the Foreign Exchange market. Use the concept of money market equilibrium, to explain what will happen to Costa Rican interest rates and the Colón/USD exchange rate in the **short-run** and illustrate on your picture. Hold the expected future exchange rate constant.